AVIAN NEUROLOGICAL DISORDERS

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1. ANATOMY OF THE NERVOUS SYSTEM

The avian nervous system presents many similarities with the mammalian nervous system but presents also many differences, some may be of great clinical relevance. For instance, the brain is lissencephalic, without convolutions, with an underdeveloped cerebral cortex and a well-developed *corpus striatum* which is considered the center for association in birds. The cranial nerves of birds correspond to those of mammals, with some minor differences. The spinal cord in birds is as long as the neural canal without *cauda equina*. Birds also have two spinal cord enlargements, the cervical and the lumbosacral enlargements, as well as a unique structure in the lumbosacral spinal cord between the dorsal columns called glycogen body. The spinal cord has an internal vertebral plexus that runs the entire length of the vertebral column. The brachial plexus is formed by ventral branches of a variable number of spinal nerves. There are three nerve plexuses in the lumbosacral region: lumbar, ischiatic and pudendal. These nerve roots lie embedded in the fovea of the synsacrum and are surrounded by the kidneys.

2. ETIOLOGY AND PATHOPHYSIOLOGY OF NEUROLOGIC DISORDERS

Neurologic disorders are common in birds and may affect the central nervous system (CNS), which includes the brain and spinal cord, and the peripheral nervous system (PNS). The PNS includes cranial nerves, spinal cord nerve roots, spinal nerves, peripheral nerve branches and the neuromuscular junction. Seizures can result from primary or secondary disorders of the brain that cause spontaneous depolarization of cerebral neurons. Seizures have three components: aura, ictus and postictal phase. Ataxia results from disorders that interfere with the recognition or coordination of position changes involving the head, trunk, or limbs. Ataxia is divided in three categories: sensory, vestibular, and cerebellar. Paralysis and paresis result from disorders that cause motor deficits. Paresis may be presented with upper motor neuron clinical signs (loss of voluntary function, normal or increased spinal reflexes, increased tone) when affects the cerebrum, brain stem or spinal cord, or with lower motor neuron clinical signs (loss of voluntary function, loss of reflexes, and weakness and muscle atrophy) when affects peripheral nerves. Head tilt, circling, or nystagmus can be caused by central (brain stem) or peripheral (middle or inner ear) vestibular disease. Peripheral lesions cause head tilt toward the side of the lesion, and usually the bird falls or circles towards the side of the lesion. Central lesions may cause head tilt, or circling in the opposite direction.

The different etiologies for avian neurologic disorders include: vascular, inflammatory/ infectious, traumatic, toxic, anomalous (congenital), metabolic, idiopathic, nutritional, neoplastic, and degenerative (VITAMIND). The neurologic disorders might involve the central nervous system and or the peripheral nervous system.

Vascular

Neurological signs due to ischemic and hemorrhagic cerebral infarctions have been reported in birds and in most instances secondary to atherosclerosis [1,2]. The pathophysiology of a brain infarction is based on the principle that the brain relies on a constant supply of glucose and oxygen to maintain ionic pump function. When perfusion pressure falls to critical levels, ischemia results in infarction. With cerebral infaction, neurologic deficits develop abruptly and depend on the location of the vascular insult and can result in seizures, vestibular signs, motor deficit, cognition impairment and monocular vision loss. Intravascular cartilaginous emboli in the spinal cord of turkeys has also been reported, and attributed to handling in animals with preexisting cartilage abnormalities [3]. In some of these cases the turkeys developed myelomalacia or spinal cord necrosis consistent with thrombosis and resultant ischemia, while others recover after a period of paresis and ataxia.

Inflammatory

Inflammatory disease of the CNS can be non-infectious or infectious. Non-infectious diseases that can cause inflammatory lesions include some toxins, autoimmune disease, and immune-mediated conditions, none of which have been well documented in birds [4]. Viruses, bacteria, fungi, protozoa, and metazoan parasites all can cause inflammatory disease of the CNS. Bacterial organism, including *Salmonella* spp., *Pasteurella* spp., *Klebsiella* spp., *Pseudomonas* spp., *Streptococcus* spp., *Enterococcus* spp., *Escherichia coli, Coxiella* spp., *Listeria* spp., *Staphylococcus* spp., *Chlamydia psittaci, Mycobacterium* spp. can cause meningitis, encephalitis, and/or myelitis. The infection of the brain and meninges can be from direct extension of an infection from the sinuses, nasal cavity, or inner ear or may be the result of a bacteremia or septicemia [4]. Bacterial granulomas may compress or invade the central or peripheral nervous system. Bacterial discospondylitis resulting in neurological deficits caused by *Staphylococcus aureus* has been diagnosed in a penguin [5], and suspected to be caused by *Corynebacterium amycolatum* in another [6]. Fungal organism like *Aspergillus* spp., *Crytococcus* spp., *Mucor* spp., may cause inflammation in the nervous system. Fungal granulomas may compress or invade peripheral nerves and cause unilateral or bilateral paresis or paralysis [7]. Parasites, such as *Baylisascaris procyonis, Toxoplasma gondii, Sarcocystis* spp*., Schistosomas* spp*., Chandlerella, quiscali, Leucocytozoon* spp., have been reported to cause neurologic disease in birds. Avian bornavirus, West Nile virus, avian paramyxovirus, avian influenza virus, togaviruses (Eastern and Western equine encephalitis), avian picornavirus, avian polyomavirus, gallid herpesvirus 2 and avian adenovirus are known to affect the nervous system of birds.

Toxic

Lead toxicity is one of the most common causes of toxicity in birds especially in waterfowl, raptors and companion birds. Lead toxicity causes encephalopathy as a result of diffuse perivascular edema, increases in cerebrospinal fluid and necrosis of nerve cells. Peripheral neuropahty is caused by demyelination of nerves and block of the presynaptic transmission by competitive inhibition of calcium. The pathogenesis of lead toxicity in the nervous system have been studied in mallard ducks [8]. Zinc toxicity could result in neurological signs, but lesions in the nervous system are rarely described. Organosphosphates and carbamates are cholinesterase inhibitors that bind to and subsequently inactivate acetycholinesterase, causing an accumulation of acetylcholine at the postsynaptic receptors. Organophosphate bonds are considered irreversible while carbamate bonds are slowly reversible over several days. Two types of neuropathy and corresponding clinical signs have been described related to toxicosis with acetylcholinesterase inhibitors. Acutely, clinical signs are related to excessive stimulation of acetylcholine receptors. Signs include weakness, ataxia, wing twitching and muscle tremors, opisthotonos, seizures and prolapse of the nictitans [9]. The second type of neuropathy is an organophosphate ester-induced neuropathy, which is not associated with an inhibition of acetylcholine. The onset of clinical signs is delayed (7 to 21 days after exposure) and is the result of a symmetric distal primary axonal degeneration of the central and peripheral nervous systems, with secondary myelin degeneration [9]. *Clostridium botulinum* toxins, most common type C but occasionally types A and type E, are a frequent cause of neurological disorders in aquatic birds [10], and have been reported rarely in other group of birds [11]. The toxins interferes with the release of acetylcholine at the motor endplate causing signs of peripheral neuropathy. Most birds exhibit hind limb paresis first, which progresses to paralysis of the wings, followed by loss of control of the neck and head in the terminal stages. Mycotoxins, such as trichothecens, have also been reported to cause neurological signs in chickens [12] and cranes, ([13], although primary histopathological lesions in the nervous system are not reported. Algal biotoxins, such as saxitoxins, brevetoxins, domoic acid and cyanotoxins, are neurotoxins that have been associated with mass mortalities of aquatic birds [14]. Saxitoxins block sodium channels, preventing signal transmission along nerves. Domoic acid is a neuroexcitatory aminoacid analog of L-glutamate produced by diatoms and some algae. Domoic acid binds to glutamate receptors in the central nervous system, causing sustained membrane depolarization and leading to neuronal excitotoxic death. Elevation of endogenous glutamate potentiates the process. Neurons in the limbic system, including the hippocampus, are most at risk. Plants have also been reported to cause neurological signs in birds. Crown vetch (*Coronilla varia*) was found to be poisonous to a budgerigar that ingested leaves from a plant next to its cage [15]. Toxicity is due to nitroglycoside, a chemical that may affect the nervous system and can cause formation of methemoglobin. Some *Kalanchoe* species have also been documented to cause neurotoxicity in chickens in an experimental study. The chickens showed depression, ataxia, muscle tremors, seizures, paralysis, and death [16]. Avian vacuolar myelinopathy (AVM) affects bald eagles (*Haliaeetus leucocephalus*), American coots (*Fulica americana*), waterfowl, and other birds in the southeastern United States resulting in severe vacuolation of white matter of the brain. The cause of the disease thought to be a naturally produced toxin produced by a cyanobacteria associated with aquatic macrophytes, most frequently hydrilla (*Hydrilla verticillata*) [17]. Tick paralysis is caused by neurotoxins of several species of hard and soft ticks, and results in a motor polyneuropathy characterized by a progressive, ascending, flaccid motor paralysis [18;,19]. Clinical signs include ataxia, paresis, paralysis, areflexia, hypotonus and respiratory failure and death if ticks are not removed. Iatrogenic toxicities have also been reported in birds, including numerous drugs like dimetridazole, chlorpyrifos, ivermectin, chlorequine and others.

Trauma

Head trauma is common in wild and companion birds. Causes of head trauma in companion birds include flights into windows, ceiling fans, and mirrors, as well as bites from other animals and other crush injuries, while in wild birds include flights into cars, wind turbine and windows, shooting, trapping and predatory wounds. Traumatic brain injury can be primary, from direct tissue damage to the brain (e.g. lacerations, contusions, hemorrhage) with or without skull fractures, or secondary from compromised blood flow to the tissue that can lead to ischemia and metabolic derangements. Brain injuries described in birds flying into towers and windows are damage to cerebellum, blood vessel rupture, herniation of the cerebellum and medulla through the *foramen magnum*, extensive subdural bleeding, and intracranial edema [20]. Spinal cord trauma can result from vertebral fractures [21], subluxations [22], luxations and rupture intervertebral disks [23]. Spinal cord injury can also be primary, as a direct damage to the tissue, or secondary caused by reduced tissue perfusion, microvascular damage, thrombus formation, and vasospasm, which leads to necrosis of the compromised tissue. Peripheral nerve trauma occurs as well in birds, with different degrees of nerve injury. Bone fractures or penetrating wounds may be associated with peripheral nerve injuries, causing unilateral peripheral neuropathies. Brachial plexus avulsion from trauma occurs most commonly in wild birds with denervation of the affected wing that results in lack of pain perception, paralysis and muscle atrophy of the wing [24]. Trauma resulting in damage to the sympathetic innervation of the eye, which includes one upper motor neuron that originates in the hypothalamus and two lower motor neurons that originate in the thoracolumbar spinal cord and the cranial cervical ganglion, causes Horner's syndrome in birds [25].

Anomalous (congenital).

Hydrocephalus has been reported in birds, and usually involves the lateral ventricles, which are grossly distended, and leads to compression of the overlying cortex [4]. While it appears that in some cases might be a congenital lesion, it seems also to be reported in older birds indicating that could possibly be acquired [26-29]. Hemivertebrae, a type of congenital vertebral anomaly and results from a lack of formation of one half of a vertebral body, was suspected in a 10-week-old African black-footed penguin (*Spheniscus demersus*). Lafora body disease is a presumed inherited defect of carbohydrate metabolism that causes the formation of typical polysaccharide complexes called Lafora-bodies. It has been reported in a cockatiel (*Nymphicus hollandicus*) [30]. In chickens, an autosomal recessive mutation has resulted in high susceptibility to seizures, and has been used for extensive research as a model for mammalian epilepsy [31].

Metabolic

Disorders of sodium and osmolality produce CNS neuronal depression, with encephalopathy as the major clinical manifestation; these disorders can also provoke CNS neuronal irritability. Neurological signs have been reported in a conure after eating salted mixed nuts [32], as well as in house sparrows (*Passer domesticus*) that ingest small numbers of road salt granules [33]. Conversely, hypocalcemia [34] and hypomagnesemia [35] cause mainly CNS neuronal irritability with seizures. In contrast, disorders of potassium rarely produce clinical signs in the CNS but may be associated with muscle weakness as the major clinical manifestation. Hypoglycemia is another metabolic cause of seizures, especially in young weanling birds, but also in birds with food depravation, endocrine and liver disease. Hepatic encephalopathy has been reported in birds with compromised hepatic function like hepatic lipidosis, hemochromatosis [36], hepatic neoplasia and other hepathopaties [9], while there are no reports documenting it with hepatic shunts. Nitrogenous waste products, including ammonia, accumulate in the blood crossing the blood-brain barrier and resulting in toxicity in the CNS.

Idiopathic

Idiopathic epilepsy is a diagnosis of exclusion, and may be applied when all other possible causes of seizure activity have been ruled out and the bird is normal between seizure events.

Nutritional

Hypovitaminosis B1 (thiamine), B2 (riboflavin), B6 (pyridoxine) and B12 (cyanocobalamine) have been reported to cause neuropathies. Hypovitaminosis B1 in birds has been studied experimentally in several species including pigeons [37], quails [38], and chickens [39]. There are also cases reported in the literature affecting raptors [40;,41], and honeyeater (Holz and Phelan, 2000). Thiamine deficiency it is also a concern in piscivorous birds fed fish with thiaminases. The lesions in the nervous system are variable depending the species, age and chronicity of the deficiency, and could affect the CNS causing poliencephalomalacia and peripheral nerve myelin degeneration. Hypovitaminosis B2 has been extensively studied in chickens [42] and pigeons [43] and leads to myelin degeneration of the peripheral nerves, resulting in clinical signs that include toes curled outward, and leg and wing paralysis. Hypovitaminosis B6 in chickens [44] has also been studied. The affected birds showed neurological clinical signs that included ataxia, head tilt and death without histopathological abnormalities in the nervous system. Vitamin E deficiency can cause encephalomalacia that results in tremors, ataxia, head tilt, cycling and/or recumbency and it is mostly a disease of captive piscivorous birds [45], but has been reported in chickens (*Gallus g. domesticus*) [46], turkeys (*Meleagris gallopavo*) [47], emus (*Dromaius novaehollandiae*) [48], raptors [45], and rarely in psittacines [4].

Neoplastic

Primary neoplasia of the nervous system reported in birds include astrocytomas, glioblastomas, oligodendrogliomas, choroid plexus papillomas, neuroblastoma, ganglioneuromas, meningiomas, pituitary neoplasia and peripheral nerve sheath tumor [4,9]. Pituitary neoplasia has more frequently been reported in middle to young age budgerigars [49], but also in other species [50]. The majority of the pituitary tumors in the budgerigars were invasive and some metastasized. Clinical signs reported include, ataxia, seizure like activity, abnormal posture, blindness, head tilt and circling. Peripheral nerve sheath tumor has been reported in a golden eagle (*Aquila chrysaetos*) with tetraplegia [51]. Metastatic neoplasia affecting the nervous system reported in birds including brain carcinoma, lymphosarcoma, hemangiosarcoma, and malignant melanoma have also been reported [4]. Pulmonary neoplasia in birds have been reported to invade the vertebrae causing compression of the spinal cord and hind limb paresis or paralysis [52, 53]. Renal neoplasia, most commonly carcinomas in budgerigars, will cause sacral nerve plexus compression and unilateral limb paresis or paralysis [54].

Degenerative

Lysososmal storage disease, a wide group of disorders characterized by the accumulation of macromolecules in the lysosomes of various cell, has been described in emus with severe, progressive and eventually fatal gangliosidosis in emus [55] and more recently in an African grey parrot (*Psittacus erithacus*). Cerebellar degeneration of unknown cause was seen in a turquoise parrot (*Neophema pulchella*). No gross change was noted, but histologically there was neuronophagia and necrosis [4]. Mineralization is an occasional incidental finding in the brain, as is lipofuscin deposition in neurons. Lipofuscin is a common finding in the neural cell bodies of older parrots, and it is suspected that in most cases this pigment accumulation does not have a functional significance [4].

3. HISTORY AND NEUROLOGIC EXAMINATION

In suspected neurological disease cases, it is necessary to determine if a neurological lesion is causing the disease, then the location of the lesion and extent, the pathologic process, the prognosis, and finally the treatment. A complete history and physical examination should be performed. Species, age and sex of the patient provide important clues to the diagnosis. Budgerigars (*Melopsittacus undulatus*) are more likely to develop renal neoplasia that can cause unilateral limb paresis by compressing the sciatic nerve. African grey parrots are commonly presented for seizures or ataxia associated with hypocalcemia. Female birds are more likely to develop vascular diseases (e.g. atherosclerosis) that could results in disorders of the neurological system. Young birds are more likely to have congenital disorders and infectious disease. Older animals are likely to have vascular, neoplastic or degenerative diseases. The onset, course, duration, and symmetry of the neurological problem also provides important information. Acute presentations are more likely to be associated to vascular, infectious, traumatic and toxic etiologies, while chronic presentations are more likely associated to metabolic, neoplastic or degenerative. The husbandry may provide information regarding possible infectious, traumatic, toxic, trauma, metabolic or nutritional etiologies. Birds housed outdoors in areas of risk are more likely to be exposure to infectious etiologies such as West Nile Virus, Sarcocystis or Baylisascaris. Female cockatoos housed with males might be bitten in the neck by their mates resulting in cervical vertebrae fracture. Birds recently exposed to other birds may become ill because of infectious diseases like proventricular dilation disease. Psittacine birds that are in galvanized cages or with toys made of unknown metals are more likely to develop heavy metal toxicity. Birds with deficient or unbalanced diets in minerals, vitamins and fatty acids are more likely to develop metabolic or nutritional diseases that affect the neurological system. Specific history of neurological disease may include a period of disorientation followed by loss of grip in the perch with subsequent tremoring and extension of the legs and wings or rigidity for varying length of time; limb weakness with loss or retention of some voluntary movement and deep pain; uncoordinated movements of the head or limbs; head tilt with or without falls, drifts or circles towards on side; impaired vision or hearing. Non-specific signs of disease are also frequently present such as lethargy and anorexia.

The physical examination may reveal findings like seizures, paresis, paralysis, ataxia, head tilt, nystagmus, intention tremor, dysmetria, visual or hearing deficits that should be evaluated in further detail during the neurologic examination. Non-specific findings like low body condition, integumentary or musculoskeletal system abnormalities and gastrointestinal or respiratory signs might add information regarding possible etiology. Birds with trauma may have blood in mouth, ears or eyes, or bruises or fractures. Horner's syndrome in birds is presented consistently with ptosis, and less often with miosis, and more rarely erection of the neck and head feathers, and is the result of loss of sympathetic innervation to the eye from a central or peripheral lesion [25]. Birds with articular gout or severe arthritis can have abnormal gait mimicking neurologic disease. Psittacine birds with avian ganglioneuritis may be seen with central nervous system signs with or without gastrointestinal sings and low body condition. Vomiting, diarrhea and/or hematuria might be seen in birds with heavy metal toxicity, but also with common bacterial and viral infectious diseases. Respiratory signs may accompany neurological sings in bacterial, viral and fungal infectious diseases, but also respiratory toxicities.

The goal of the neurological examination is to confirm a neurological abnormality and to localize the abnormality within the nervous system. The neurological examination includes the following parts: observation, palpation, functional assessment of the cranial nerves, evaluation of proprioceptive deficits and spinal reflexes, and sensory evaluation.[56] The neurological examination in birds have only been evaluated in pigeons (*Columba livia domestica*), mute swans (*Cygnus olor*), common buzzards (*Buteo buteo*), common kestrels (*Falco tinnunculus*) and northern goshawks (*Accipiter gentilis*).[57]

Observation of a bird's behavior provides essential information about its neurologic function. Changes in **mentation**, such as depression, lethargy, stupor, or coma, can signal systemic illness or central nervous system (CNS) damage. A modified Glasgow Coma Scale (MGCS) for use in raptors presenting with head trauma exists.[58] **Posture** depends on CNS and peripheral nervous system (PNS) coordination; abnormalities like **recumbency**, **opisthotonus**, or **rigidity** may point to brainstem or cerebellar lesions. **Attitude**, or the positioning of the head, eyes, and limbs, is primarily controlled by the **vestibular system**, which has both central and peripheral components. Dysfunction may cause **nystagmus**, **head tilt**, **falling**, or **circling**, often toward the side of the lesion. A **wide-based stance** on a perch is common in vestibular disorders, ataxia, or weakness. **Movement and gait** reflect coordination between the cerebrum, cerebellum, and vestibular system. Birds with **cerebral dysfunction** may walk but have difficulty navigating perches. **Cerebellar disease** can cause **intention tremors** (e.g., head shaking while eating) and dysmetric limb movements. **Vestibular dysfunction** leads to **balance loss** and **circling**, while **paresis or paralysis** may result from damage to motor pathways from the brain to the limbs. **Seizures** are common signs of cerebral dysfunction and may also result from **toxins**, **parasites**, **metabolic issues**, or **nutritional deficiencies**.

The **cranial nerve (CN) examination** is critical in birds for localizing neurologic lesions. Differences from mammals exist, such as **striated muscle in the iris**, giving birds some **voluntary pupil control**, and **nictitating membrane control by CN VI**. **CN II (optic)** is assessed via **obstacle avoidance** and **menace response**, though this can be absent in birds. The **PLR** may be complicated by incomplete bony septa between eyes, allowing light to stimulate both retinas. **Eye deviation** can help localize CN **III**, IV, or VI dysfunction. **CN V (trigeminal)** provides facial sensation and controls the beak and eyelids; dysfunction may cause a **dropped beak or chewing problems**. **CN VII (facial)** contributes to jaw movement and taste, though facial muscles are minimal in birds. **CN VIII (vestibulocochlear)** plays a key role in **balance**. Peripheral vestibular lesions cause **spontaneous horizontal nystagmus**, **head tilt**, and **circling**, while **central lesions** also affect **mentation and proprioception**. The final cranial nerves—**CN IX, X, XI, and XII**—have overlapping roles in **swallowing**, **tongue control**, and **autonomic functions**. **Tongue deviation** and **reduced tone** suggest **CN XII injury**, while **dysphagia or regurgitation** may indicate **vagal or glossopharyngeal nerve** damage.

Postural reactions in birds assess the integrity of both sensory and motor pathways and are useful in detecting subtle neurologic deficits. These reactions involve complex input from the **peripheral nervous system (PNS)** to adjust muscle tone and the **central nervous system (CNS)** to coordinate responses. Tests include placing a wing or leg in an abnormal position and observing the bird's ability to correct it. **Conscious proprioception**, or limb awareness without vision, is tested by placing the foot dorsally on a perch or sliding a sheet under it—failure to reposition the foot indicates deficits. Birds standing on knuckles show proprioceptive loss. **Wing correction** and **hopping reactions** further test coordination, but many birds resist these due to stress. The **drop and flap test**, simulating a fall, evaluates thoracic limb response, with asymmetry or delay suggesting CNS lesions. **Extensor postural thrust** assesses pelvic limb function when the bird is lowered toward the ground, simulating a walking motion.

Spinal reflexes evaluate **segmental reflex arcs** independently of the brain. Birds show species variation in spinal nerve number (e.g., 38 in pigeons, 51 in ostriches), and their **brachial** and **lumbosacral plexuses** serve the wings and legs respectively. The **vent sphincter reflex** assesses **pudendal plexus** integrity—normal response is a wink-like contraction and tail bob. **Loss of tone** suggests lower motor neuron (LMN) damage; hypertonicity suggests upper motor neuron (UMN) disease. The **pedal flexor reflex** tests the **ischiatic nerve** via foot pinch; absence suggests LMN damage or lumbosacral compression (e.g., renal mass or egg impingement). A **positive crossed extensor reflex**—extension of the opposite limb during testing—indicates UMN disease cranial to the lesion. The **patellar reflex**, though sometimes hard to elicit due to anatomy or restraint, assesses the **femoral nerve**. **Wing withdrawal**, tested by pinching the major digit, assesses the **brachial plexus**. As in the legs, a **positive crossed wing reflex** suggests cervical spinal cord or brainstem disease.

Cutaneous sensation is assessed last to avoid stress and tests **pain perception** and sensory integrity. Birds lack a panniculus reflex but have **sensory innervation to feather follicles**. Plucking or pinching feathers can help localize lesions. Because avian dermatomes are not mapped, **nociceptive testing** begins caudally and progresses to stronger stimuli. **Conscious perception**—indicated by vocalizing, turning, or escape—confirms intact sensory pathways to the brain. Importantly, a reflex withdrawal can still occur even if the spinal cord is transected cranial to the reflex arc.

A modified Glasgow Coma Scale (MGCS) for use in raptors presenting with head trauma exists.[58] The avian MGCS was developed with 3 assessment components (motor activity, level of consciousness, and brain stem reflexes), each with a scale from 1 to 5, creating a score ranging from 3 to 15

4. DIAGNOSTIC TESTING

Complete blood count, plasma chemistry panel and survey radiography are the minimum data base required when diagnosing neurological disorders. Ancillary neurodiagnostic tests include cerebrospinal fluid analysis, myelography, computed tomography, magnetic resonance imaging, scintigraphy and electromyography. Additional tests to detect infectious disease agent or antibodies, as well as toxin screen might be indicated based on the differential diagnoses.

Survey radiography should be used to evaluate the skull, axial skeleton, and spinal column, particularly in cases of trauma. Sedation or general anesthesia facilitates exact positioning to yield the best image of the desired area. Orthogonal projections or three views in the case of skull should be obtained with the beam centered and collimated on the area of interest.

Cerebrospinal fluid analysis is most useful in characterizing infectious, neoplastic or inflammatory disorders of the spinal cord or brain but is not performed routinely in birds. The subarachnoid space in birds is narrow, so that fluid collection from the cerebromedullary cistern is difficult. A proliferative network of blood vessels that participates in cerebrospinal production (choroid plexus) lies paramedially bilaterally in the caudomedial aspect of the fourth ventricle. Large venous sinuses lie laterally on the interior surface of the occipital bones of the cranial cavity. The collection site must be approached at midline, or significant hemorrhage may occur. The atlanto–occipital joint is flexed at a 30-degree angle in a well-aligned laterally recumbent patient, a 25 to 27 gauge needle with a translucent hub is placed at dorsal midline slightly caudal to the occipital protuberance, directed rostrally at a 45-degree angle to the horizontal axis of the head, and advanced slowly through the skin at 1 mm intervals. The volume that can be collected ranges from 0.1-0.5 mL. The technique has been described in psittacine birds [59]. The author reported CSF of similar composition to mammals and mild deleterious effects in the animals.

Myelography has been used rarely in clinical avian practice. Its limited use in birds is likely because of rare clinical indications, lack of familiarity with the technique, fear of iatrogenic injury to the spinal cord, and concern about potentially fatal complications of the procedure. The cerebellomedullary cistern is very small in many avian species and directly overlies a large venous plexus. It is considered by some authors that contrast medium into the subarachnoid space in the cerebellomedullary cistern cannot be consistently repeated and trauma to the spinal cord near the brain stem can result in death. The technique has been described in pigeons using the atlanto–occipital space for insertion of a 26 gauge 2.5 cm needle to deliver 0.2 mL iohexol over 1 minute, followed by elevation of the head for 5 minutes [60]. Post-myelographic complications were not observed and the quality of the myelogram was considered acceptable. However, the cistern is relatively larger in pigeons than in some other avian species and accommodates a needle for injection of contrast medium. The fused vertebrae of the synsacrum, the glycogen body, and the absence of a *cauda equina* interfere with the typical mammalian technique of lumbosacral puncture for subarachnoid access necessitating for a thoracolumbar approach. The technique has been evaluated in chickens using a 25 gauge, 4 cm spinal needle at the site followed by administration 0.8

to 1.2 mL/kg iohexol contrast medium into the subarachnoid space at 0.5 mL [61]. The technique showed some promised but the authors only were successful in 3 out of 6 animals, being unable to inject in the subarachnoid space or resulting in complications that lead to death in the other 3 cases.

Computed tomography (CT) imaging is best indicated to evaluate abnormalities in the skeletal structures and respiratory tract but has also been used in birds with neurologic disease due large intracranial lesions or large spinal cord lesion; for other brain or spinal cord diseases additional techniques, such as magnetic resonance imaging (MRI), may be necessary [62]. Soft tissues structures are better visualized with contrast administration, especially where there is increased blood flow. A dose of 2.22 mg/kg of iodinated contrast material IV is recommended [56]. Examples of CNS disease diagnosed with the aid of CT imaging in birds include hydrocephalus, peripheral vestibular disease due to otitis media [63], and spinal cord compression due to vertebral invasion of bronchogenic carcinoma [64], intervertebral disc rupture [23], dysconpodylitis [65], and vertebral fracture. The limitations to the use of CT for imaging structures and organs are mainly dependent on the spatial resolution of the CT scanner and size of the patient.

Magnetic resonance imaging (MRI) is best indicated to evaluate soft-tissue and has been found very useful for visualizing and evaluating various parts of the CNS, including the cerebral hemispheres, cerebellum, optic chiasm, brain stem and spinal cord. Gadolinium, a paramagnetic intravenous contrast agent can be used as a contrast medium for MRI studies (dose 0.25 mmol/kg). After administration of the compound via an IV catheter the gadolinium is distributed throughout the body, and changes the local magnetic field in tissues in which it is present in high concentrations (i.e. highly vascularized tissues), allowing delineation from the surrounding structures. Examples of diseases of the nervous system diagnosed with the aid of MRI in birds include ischemic infarct [66] and hemorrhagic infarct [1], hydrocephalus [67,68], peripheral vestibular disease due to otitis media [63], spinal cord trauma[69], peripheral nerve sheath neoplasia [51], spinal cord compression due to hemivertebrae [70], and intervertebral disc rupture [23].

Nuclear imaging or scintigraphy is a non-invasive method for evaluation of soft tissue and osseous structures associated with the nervous system. The technique involves intravenous administration of a small amount of a gamma-emitting radionuclide alone (e.g. technetium-99m (99mTC) that will allow it to accumulate in certain tissues. A gamma camera is used to record the amount of radiation emitted from the body and to create images of the distribution of the radionuclide throughout the patient's body. This technique has been found useful to identify spinal abnormalities in birds [71].

Electromyography, nerve conduction studies, and muscle biopsy are used to help differentiate transmission disorder, neuropathy, and myopathy in neuromuscular disease. Electromyography (EMG) is the study of the electrical activity of muscle by insertion of a recording electrode into the muscle. It examines the integrity of the motor unit, which consists of the lower motor neuron and the muscle fibers that it innervates. Normal resting muscle does not show observable electrical activity once the electrode placement is stabilized and no audible signal is created. Electrical activity demonstrated can be separated into three categories: insertional (associated with electrode movement), spontaneous (associated with resting muscle) and evoked (associated with electrical stimulation of nerves). Electromyography has been shown to be useful in birds in the diagnostic of traumatic brachial plexus injury [24] and limb denervation [72]. Nerve conduction studies have become a simple and reliable test of peripheral nerve function. The conduction velocity is measured between the two stimulus points on the nerve, thereby eliminating the time for neuromuscular transmission and generation of muscle action potential. It is derived as the ratio between the distance between two stimulation cathodes and the corresponding latency difference. Reference ranges for the motor nerve conduction velocities have been established in barred owls and rheas [73], and mallard ducks [74]. Nerve conduction studies have been used in the diagnosis of peripheral demyelinating neuropathy secondary to lead toxicity [75]. Muscle biopsy is indicated to confirm, define and possibly provide a cause for motor unit disease. In acute disease, a severely affected muscle is selected. With chronic disease, a muscle demonstrating only moderate changes is the best choice.

5. TREATMENT

The treatment of the different disorders of the nervous system require addressing the primary cause targeting the therapy to a specific etiology, while providing supportive care to recover from the primary lesions and preventing secondary damage.

Status epilepticus therapy is aimed on stopping the seizure activity and preventing the reoccurrence of seizures. Before starting antiepileptic therapy, a basic history from the owner must be collected with the signalment, previous history of seizures, current drug therapy, and history of trauma or exposure to toxins. Antiepileptic therapy should be started as soon as possible with benzodiazepines like midazolam (0.5-2 mg/kg IM or IV) or diazepam (0.5-1 mg/kg IV or IO). Benzodiazepines have a short duration of action needing frequent administration, and long-term usage leads to tolerance and decreased drug efficacy. Additional diagnostic test should be performed as soon as the patient is stable, and specific treatment towards the underlying cause of the seizures should be initiated. Perches should be removed from the cage, soft bedding should be provided, and food and water should be placed in shallow bowls or spread on the cage floor if necessary. The maintenance antiepileptic drugs used for many years in avian medicine are phenobarbital and potassium bromide, but information in newer drugs like levetiracetam, zonisamide, and gabapentin have recently become available in birds. In epilepsies of unknown cause, antiepileptic treatment is introduced at the 2nd convulsive epileptic seizure. An attempt at tapering off the antiepileptic drug can be done when the bird has been seizure-free more than 2 years. The tapering of medication should be done slowly over several months before discontinuation of the treatment. If the epilepsy is structural, with an obvious demonstrated lesion, the epileptic focus rarely disappears spontaneously and the antiepileptic drugs are continued indefinitely. Phenobarbital is a potent antiepileptic drug that is available in oral and parenteral forms. It is effective as an anticonvulsant in part because of its potentiating action on the inhibitory neurotransmitter gamma aminobutyric acid, which increases the seizure threshold and lowers the electrical activity of the seizure focus. The potential secondary hepatotoxicity necessitates close monitoring of liver parameters. Beside the common clinical side effects (sedation, polyuria, polydipsia, polyphagia), its activation of the hepatic microsomal enzymes (p450 system) alters the metabolism of other antiepileptic drugs metabolized by the liver such as levetiracetam and zonisamide. In some species, phenobarbital might activate of its own metabolism necessitating multiple serum level measurements in the first 6 months of use and metabolic tolerance, forcing gradual increments of the dosage to maintain control, until higher end of optimal therapeutic levels or hepatotoxicity are reached. Based on pharmacokinetic studies in African grey parrots, dosages higher than 20 mg/kg PO q 12 h are needed in psittacine birds [76]. Peak and trough serum levels should be checked 2–3 weeks after initiating therapy or changing the dose until significant therapeutic levels are reached (10–45 µg/mL) or seizures are controlled. Potassium bromide can be used alone or in conjunction with phenobarbital when phenobarbital alone doesn't work to control the seizures adequately. The mechanism of action is through completion of the bromide ions with Cl- transport, resulting in hyperpolarization. The side effects are similar to those of phenobarbital but without the potential for liver toxicity, and it does take a longer time for the body to eliminate the drug once it has been discontinued due to the long half-life. Based on clinical reports in psittacines, 80 mg/kg PO q 24 h are recommended but pharmacokinetic studies are lacking [77]. Peak and trough serum levels should be checked 2–3 weeks after initiating therapy or changing the dose until significant therapeutic levels are reached (1-3 mg/mL) or seizures are controlled, but it might take 2-3 months to reach a steady-state. Levetiracetam is a newer antiepileptic drug available in oral and parenteral forms. The mechanism of action is not entirely understood, but it has been postulated to involve inhibition of excitatory neurotransmitter release by binding to the synaptic vesicle protein SV2A, thereby modulating calcium dependent exocytosis of neurotransmitters. It also suppresses the inhibitory effect of Zn^{2+} on gammaaminobutyric acid and glycine-gated currents. Based on pharmacokinetic studies in Hispaniolan Amazon parrots, 50-100 mg/kg PO q 8-12 h are recommended in psittacines. [78]. Zonisamide prevents seizure activity through several mechanisms of action including ion channel modulation, enhancement of neurotransmitters and inhibition of carbonic anhydrase. Zonisamide is partially metabolized by the liver. Based on a study in Hispaniolan Amazon parrots, zonisamide administered at 20 mg/kg PO q 12 h in psittacines is likely to achieve target plasma concentrations and be safe [79]. Other drugs like gabapentin and clonazepam have been used in clinical reports, but evidence regarding their efficacy is lacking [2]. Peak and trough serum levels should be checked 1 week after initiating therapy or changing the dose until significant therapeutic levels are reached (10 to 40 ug/mL) or seizures are controlled. Long-term monitoring requires detailed log of number seizures and duration, adjusting therapy with antiepileptic drugs as needed to control seizures.

Cerebrovascular accidents, like hemorrhagic or ischemic strokes, requires differentiation through MRI for specific medical treatment. Treatment of ischemic stroke includes thrombolytic agents, which are contraindicated in cases of hemorrhagic lesions. Surgical treatment for hemorrhagic stroke or vascular aneurysm might not be currently feasible in birds due to size limitations. Supportive treatment after a cerebral infarct or hemorrhage is aimed at maintaining adequate tissue perfusion and oxygenation and managing neurologic and non-neurologic complications.

Traumatic brain injury medical treatment is aimed at preventing or reducing the effects of secondary brain injury through maintaining adequate cerebral perfusion pressure, preventing or decreasing cerebral edema, and controlling intracranial pressure. The bird should be placed in a dark, quiet area with an environmental temperature of 23ºC (73ºF) since hyperthermia may contribute to greater posttraumatic brain damage. Elevation of the head (up to 30º) may decrease intracranial pressure by facilitating venous and cerebrospinal fluid drainage. The administration of oxygen in an oxygen cage is recommended for moderate to severe cases. Slow administration over 15 minutes of mannitol (0.25-1 mg/kg IV) given every 4–6 hours as needed for up to 3 boluses should be considered, together with crystalloid fluids at 50% of the normal rate. Mannitol is contraindicated in hypovolemic patients. A clinical response to mannitol therapy may be seen within 5-10 minutes, and an improvement in mentation, posture, and pupillary signs indicate a positive response to mannitol therapy. If there is an insufficient response to the first dose of mannitol, an additional bolus can be given 40-60 minutes later. Lack of improvement with this therapy indicates that the brain damage will not be responsive to mannitol therapy and further therapy may not be warranted. The benefits that mannitol have on lowering intracraneal pressure in the rest of the brain are likely to outweigh the concern that mannitol may result in worsening of intracranial hemorrhage. The side effects associated with repeated mannitol administration include hyperosmolarity, hypovolemia, and electrolyte disturbances (elevated sodium most commonly), therefore, adequate fluid therapy and monitoring are necessary. Hypertonic saline (7-7.8% NaCl) can be used instead of mannitol in hypovolemic or euvolemic patients to decrease intracranial pressure 3-5 mL/kg over 15 minutes, followed by crystalloid fluids at 50% of the normal rate. Hypertonic saline is contraindicated in hyponatremic patients. Slow administration of opioid drugs are recommended to treat pain and to avoid intracranial pressure elevation secondary to the sympathetic response to pain, but caution is warranted for patients unable to adequately ventilate. The use of non-steroidal anti-inflammatory drugs in cases of traumatic brain injury in birds is only recommended once the patient is stabilized. Corticosteroids are not recommended for the treatment to traumatic brain injury. Using the avian modified Glasgow coma scale, raptors with a total MGCS of < 10 were less likely to survive than those with a score > 12.

Spinal cord trauma, like vertebral fractures, subluxations and luxations, medical treatment is aimed to prevent further primary injury by careful handling and strict restriction of activity, and to prevent secondary injury to the spinal cord [80]. If needed, sedation or general anesthesia is recommended to facilitate handling and restrict activity. Intravenous fluid therapy is recommended to improve tissue perfusion and oxygenation. Opioid drugs are recommended to treat pain secondary to the trauma. Non-steroidal anti-inflammatories, gastrointestinal protectants and nutritional support should be considered. Corticosteroids such as methylprednisolone sodium succinate have been used historically in the treatment of spinal cord injuries for their anti-inflammatory and free radical scavenging properties. However, corticosteroids can cause severe side effects including hemorrhage, gastrointestinal ulceration, and immunosuppression in birds and are not currently recommended in most cases. Surgical stabilization is indicated for an unstable spinal fracture or subluxation with decompensating neurologic signs. Physical therapy consultation regarding an appropriate therapeutic treatment has the potential to expedite recovery and improve range of motion, muscle strength, proprioception, gait, and endurance, as well as correct movement dysfunction, and decrease pain and inflammation. Physical therapy consists of examining and evaluating patient functional limitations, impairments, and disability, and may consist of active assisted range of motion, passive range of motion, active range of motion, stretching, isometric, isotonic, and isokinetic strengthening, proprioceptive exercises, core stabilization, soft tissue massage, and other therapies. Any physical therapy routine should be tailored to the specific patient because all routines are not appropriate for all injury types or individuals and treatment is adjusted based on patient response.

Bacterial CNS infections are treated with antibiotics with good CNS penetration like chloramphenicol, trimethoprim-sulfa, metronidazole, fluroquinolones and third-generation cephalosporins. Fungal CNS infection are treated with antifungals with good CNS penetration like voriconazole, fluconazole or amphotericin B. Parasitic CNS infection required identification of the organism or empirical treatment with antiparasitic drugs against the presumptive organism. Discospondylitis medical treatment consists of long-term antibiotic therapy against the causative organism or organisms cultured from the blood. Empirical therapy with amoxicillin-clavulanic or trimethoprim-sulfa with a fluoroquinolone is recommended if there is no positive culture of the lesion or blood. Vestibular disease resulting from otitis media or interna of bacterial etiology, long-term systemic antibiotics like third-generation cephalosporins and fluroquinolones, and non-steroidal anti-inflammatory therapy should be considered. Surgery with removal of the affected tissue has also been reported to treat otitis media in a goose [63].

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Table 1- Etiologic classification following the VITAMIND mnemonic of neurological diseases reported in birds

Vascular
- Atherosclerosis
- Non-traumatic cerebrovascular accidents
- Fibrocartilaginous embolism
- Lipid embolism

Inflammatory/Infectious
- Viral
  - Avian bornavirus
  - West Nile virus
  - Avian paramyxovirus
  - Avian influenza virus
  - Togaviruses (eastern and western equine encephalitis)
  - Avian picornavirus
  - Polyomavirus
  - Gallid herpesvirus 2
  - Picornavirus
  - Adenovirus
- Bacterial
  - *Salmonella* spp., *Pasteurella* spp., *Klebsiella* spp., *Pseudomonas* spp., *Streptococcus* spp., *Enterococcus* spp., *Chlamydia psittaci, Mycobacterium* spp.
  - *E. coli, Coxiella* spp. , *Listeria* spp., *Staphylococcus* spp.,
- Parasitic
  - *Baylisascaris procyonis*
  - *Toxoplasma gondii*
  - *Sarcocystis* spp.
  - *Schistosomas* spp.
  - *Chandlerella quiscali*
  - *Leucocytozoon* spp.
- Fungal
  - *Aspergillus* spp.,
  - *Crytococcus* spp.,
  - *Mucor* spp.,
- Prions

Trauma
- Traumatic brain injury with or without skull fractures
- Spinal cord trauma with vertebral fractures, subluxations, luxations, rupture intervertebral disk,
- Brachial plexus avulsion

Toxic
- Lead, zinc, copper, mercury
- Organosphosphates
- Carbamates
- Clostridium botulinum toxin
- Cyanobacteria toxin
- Domoic acid
- Salt poisoning
- Toxic plants
- Ivermectin
- Chloroquine
- Tick toxicity
- Mycotoxins

Anomalous
- Hydrocephalus
- Hemivertebrae
- Lafora body neuropathy

Metabolic
- Hypocalcemia
- Hypoglycemia
- Hepatic encephalopathy

Idiopathic
- Idiopathic epilepsy

Neoplastic/Nutritional
- Primary neoplasia of the nervous system like astrocytomas, glioblastomas, oligodendrogliomas, choroid plexus papillomas, neuroblastoma ganglioneuromas, meningiomas, pituitary adenoma,
- peripheral nerve sheath tumor
- Metastatic neoplasia of the brain carcinoma, lymphosarcoma, hemangiosarcoma, and malignant melanoma
- Pulmonary, renal or gonadal neoplasia with compression of spinal cord and sciatic nerve
- Thiamine (Vit B1) deficiency
- Riboflavin (Vit B2deficiency
- Cobalamine (Vit B12) deficiency
- Vitamin E deficiency

Degenerative
- Lysosomal storage disease

Table 2- Avian neurological examination form

Observation
Mentation: bright/alert obtunded stuporous comatose
Posture: Body: normal recumbent opisthotonus
Head: normal head tilt
Movement: Resting: normal tremor myoclonus seizures
Gait: normal paresis ataxia dysmetria circling

Palpation
Head palpation
Spinal palpation
Limb palpation LW RW LL RL

Cranial Nerves
I: Odor
II and V: Menace OS OD
II and III: Pupillary light reflex OS OD
III, IV, and VI: Strabismus OS OD
III, IV, VI and VIII: Oculocephalic reflex, OS OD
V: Palpebral reflex OS OD
V: Beak tone
VI: Nictitans movement OS OD
VIII: Head tilt L R,
Nystagmus horizontal vertical
Positional Non-positional
Fast-phase L R
IX, X, XI: Gag reflex
XII: Protrusion and retraction of the tongue

Postural Reactions
Conscious propioception: LT RT LP LP
Hopping LT RT LP RP
Placing Optic Tactile R L
Drop and flap R L

Spinal Reflexes
Perineal
Patelar L R
Leg Withdrawal L R
Wing withdrawal L R

Sensory evaluation
Feather retract

Assessment
1. Neurologically normal, abnormal
2. Neuroanatomical localization:
    Brain: Cerebrum, cerebellum, brain stem
    Spinal Cord: Cervical brachial plexus, thoracic lumbosacral plexus
    Peripheral nerve
    Neuromuscular
3. Differential diagnosis list
4. Diagnostic plan

Table 3- Cranial nerves, nerve function, applicable tests to evaluate function and clinical signs associated with dysfunction

Number and name	Function	Test	Normal response	Abnormal response
I Olfactory	Sensory– Smell	Smelling alcohol	Aversion	No reaction
II Optic	Sensory– Vision	Menace response Pupillary light reflex (PLR)	Blink or absent blink PLR present.	Absent blink PLR absent
III Oculomotor	Motor–extrinsic ocular muscles and upper eyelid muscle Parasympathetic – intrinsic ocular muscle -	Eyeball position and movement Menace response Pupillary light reflex		Ventrolateral deviation Drooped upper eyelid PLR absent with dilated pupil
IV Trochlear	Motor extrinsic ocular muscle	Eye position, eye movement	Eye centered in palpebral fissure, eye moves in all directions	Dorsolateral deviation
V Trigeminal Ophthalmic branch Maxillary branch Mandibular branch	 Sensory– upper lid, forehead skin, nasal cavity, upper beak) Sensory– both lids, hard palate, nasal cavity, lateral upper beak) Motor– orbicularis, lower lid, chewing) Sensory– lower beak skin commissures	 Response to touch Palpebral reflex Palpebral reflex Jaw palpation	 Response to touch Palpebral reflex present Palpebral reflex present Closed mouth good jaw tone	 Lack of response Unable to blink Unable to blink Unable to close jaw
VI Abducens	Motor – extrinsic ocular muscles, nictitans	Eye position, eye movement	Eye centered in palpebral fissure, nictitans moves	Medial deviation nictitans immobility
VII Facial	Motor – facial expression Sensory – taste	 Taste alcohol	Facial symmetry Aversion	Facial asymmetry Poor taste Decreased secretions

	Parasympathetic - most glands of the head			
VIII Vestibulocochlear	Sensory – hearing Sensory – balance and coordination	Startle response	Startle reaction to hanclap Oculocephalic reflex	Poor startle reaction Spontaneous nystagmus, prolonged or absent post-rotatory nystagmus. Head tilt, circling
IX Glossopharnygeal	Sensory - taste and sensation in the tongue and trachea Motor – pharynx, larynx, crop and syrinx	Response to touch and taste alcohol Gag reflex	Move tongue and aversion Swallowing	Poor taste and feel Poor gag reflex, dysphagia Voice change
X Vagus	Sensory – larynx, pharynx, viscera Motor - larynx, pharynx, esophagus, crop Parasympathetic – glands, heart and viscera	Gag reflex	Swallowing	Poor gag reflex, dysphagia Voice change
XI Spinal Accessory	Motor – superficial neck muscles	Palpate for atrophy of muscles	Normal muscles and movement	Poor muscles or movement
XII Hypoglossal	Motor – tongue, trachea, syrinx	Protrusion and retraction of tongue	Tongue protrudes symmetrically and retracts	Tongue deviates to the side, weak withdrawal

Table 4- Modified Glasgow Coma Scale (MGCS) for use in raptors presenting with head trauma[58]

Motor Activity *(check one)*
____ **5:** Normal
____ **4:** Incoordination, hemiparesis or tetraparesis (weakness on one side of both sides - wing and limb), able to stand upright onto digits and perch (if applicable)
____ **3:** Unable to perch/stand upright, incoordination and paresis (weakness)
____ **2:** Recumbent with any movement of the wings and/or limbs
____ **1:** Recumbent with increased extensor tone in wings and limbs (decerebrate rigidity) or flaccid (decreased tone) in the wings and limbs
Level of Consciousness
____ **5:** Alert, no delay in response
____ **4:** Dull or obtunded rapid response to minimal stimulus (attempt to touch)
____ **3:** Dull or obtunded/inappropriate; slow or absent response to minimal stimulus (attempt to touch), but responds to to actual touch
____ **2:** Unconscious; response to noxious stimuli (stuporous)
____ **1:** Unconscious; unresponse to noxious stimuli (comatose)
Brainstem Reflexes
____ **5:** Normal PLR, palpebral, and physiologic nystagmus
____ **4:** Bilateral unresponsive miosis, normal oculocephalic reflex, and normal palpebral
____ **3:** Pinpoint pupils (miosis) in a dark room, reduced to absent oculocephalic reflex, reduced to absent palpebral
____ **2:** Unilateral unresponsive mydriasis (anisocoria), reduced to absent oculocephalic reflex, reduced to absent palpebral
____ **1:** Bilateral, unresponsive mydriasis with reduced to absent oculocephalic reflex, reduced to absent palpebral
NEUROLOGIC SCORE TOTAL (add all 3 scores from above):

Table 5 Dosages of selected drugs used in birds.

Drug	Species	Dose	Basis
Benzodiazepine			
Midazolam	Parrots, raptors	0.5-2 mg/kg IM, IV	EU
Diazepam	Parrots, raptors	0.5-2 mg/kg IV	EU
Antiepileptic			
Phenobarbital	African grey parrot	>20 mg/kg PO q12h	PK
		2-8 mg/kg IV	EU
KBr	Umbrella cockatoo	80 mg/kg PO q24h	EU
Levetiracetam	Hispaniolan Amazon parrot	50-100 mg/kg q8-12h	PK
Zonisamide	Hispaniolan Amazon parrot	20 mg/kg PO q12h	PK

Note: PK, pharmacokinetic study; PD, pharmacodynamics study; EU, empirical use.